

TrES-5b

R-1590

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_220827 · created 2026-08-02T00:56:08+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459818.8482 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.084 +/- 0.052
Transit depth (Rp/Rs)^2	0.71 +/- 0.87 [%]
Orbital Inclination (inc)	79.88 +/- 2.88
Airmass coefficient 1 (a1)	2.44 +/- 1.8
Airmass coefficient 2 (a2)	-0.29 +/- 0.45
Scatter in the residuals of the lightcurve fit is	81.46 %
Best Comparison Star	#3 - [126, 309]
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.023 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.084 ± 0.052 vs 0.142 (deviation 0.83σ)
Duration fit vs published	0.023 d vs 0.0758 d (deviation 1.45σ)
Mid-time O–C	4.3 min
Depth SNR	1.2
Residual scatter	81.46 %

Light curve

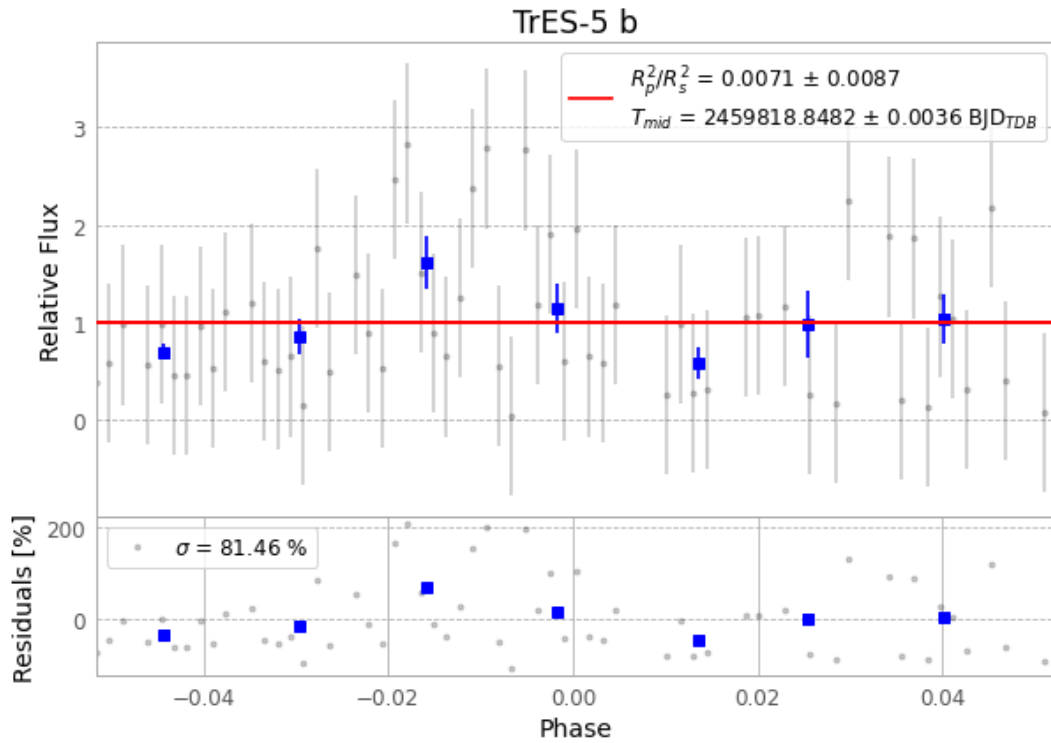


Figure 1 — FinalLightCurve_TrES-5 b_2022-08-26.png (SHA-256 c6a00bd4131a6439...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [492, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8e583b1623c098a7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), TRES-5220827062714.FITS → TRES-5220827101213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-220827.log (984be456227b...), FinalLightCurve_TrES-5 b_2022-08-26.png (c6a00bd4131a...), FinalLightCurve_TrES-5 b_2022-08-26.csv (6707ab430346...)

Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:56Z.